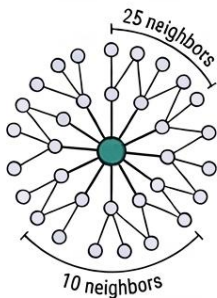


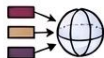


STAGE 1: Sampled subgraph

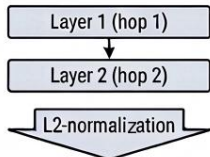


SimSample

neighbors ordered by
feature similarity, the
most similar kept



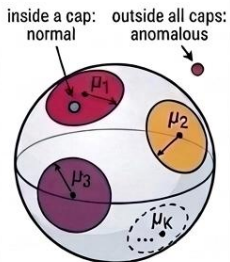
STAGE 2: GraphSAGE encoder + projection



embeddings on
the unit sphere



STAGE 3: Prototype atlas of normality

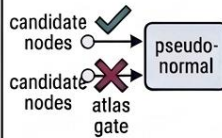
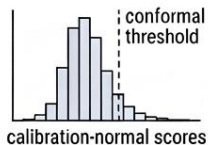


K prototypes per graph

**normality = union of
hyperspherical caps**



STAGE 4: Conformal pseudo- labeling with a geometric veto



synthetic anomalies
(mixup + halo)



STAGE 5: Score unseen anomaly classes

